AUDITING FAIRNESS IN AI-BASED RESOURCE ALLOCATION

MOTIVATION

Artificial Intelligence systems are increasingly used to allocate opportunities and resources. However, such systems may unintentionally produce biased outcomes across demographic groups. This project investigates fairness issues in AI-based resource allocation using the Adult Census Income dataset.

DATASET

- UCL Adult Census Income Dataset
- Records: 45,222
- Features: 15
- Protected Attributes: Gender, Race

WORKFLOW

DATASET → DATA CLEANING → DATA ENCODING → FAIRNESS AUDIT → PERFORMANCE EVALUATION → TRAIN MODELS → BIAS MITIGATION → RESULTS

RESEARCH QUESTION

RQ1

Does the model exhibit gender bias?

RQ2

Does the model exhibit racial bias?

RQ3

Which model performs best?

RQ4

Can fairness mitigation reduce bias?

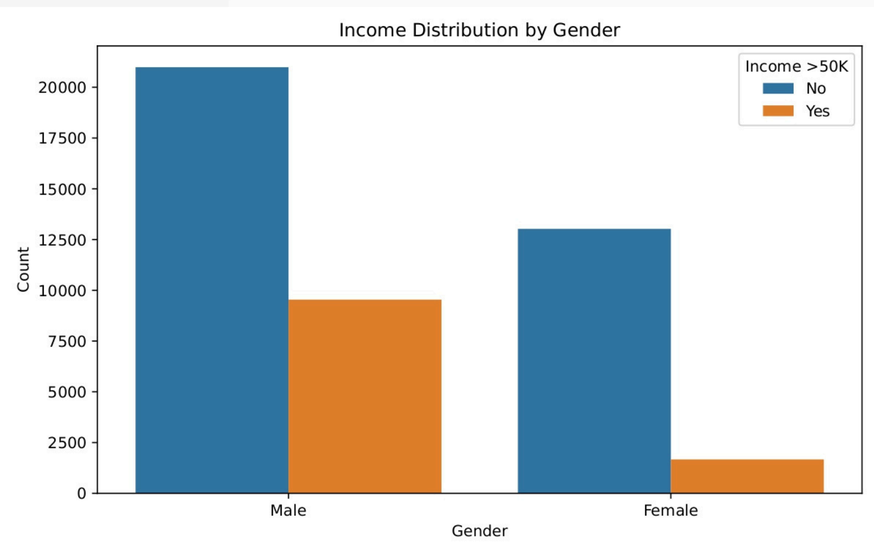
RQ5

What trade-offs exist between fairness and accuracy?

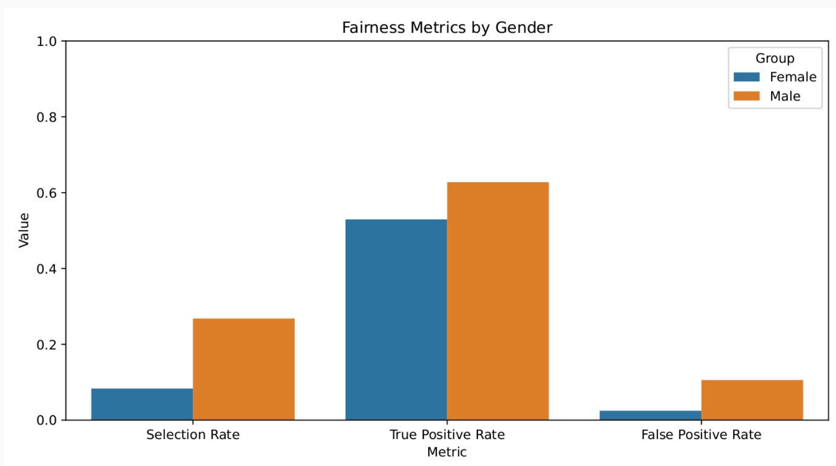
KEY RESULTS

- Random Forest achieved the highest accuracy (84.77%).
- Gender bias was detected (Female SPD = -0.184).
- Females received fewer positive outcomes than males.
- Racial disparities were observed across multiple groups.
- Black group showed the largest disadvantage (SPD = -0.138).
- Fairness mitigation reduced bias (SPD: -0.184 → -0.180).
- Accuracy remained nearly unchanged (84.77% → 84.60%).
- Fairness improved with minimal performance loss.
- AI fairness auditing revealed important demographic disparities.
- Bias mitigation can support more equitable AI decision-making.

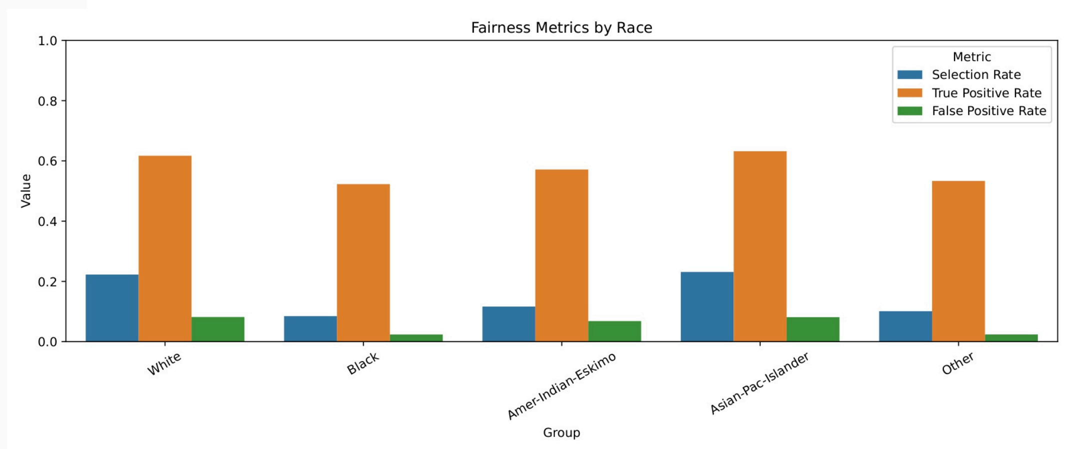
RQ1



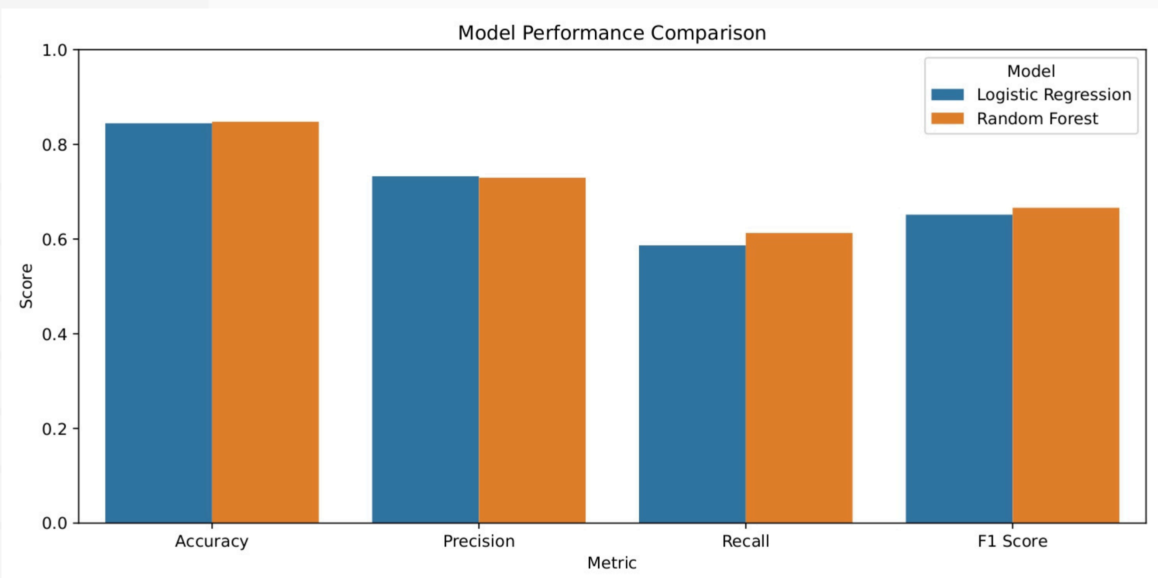
RQ3



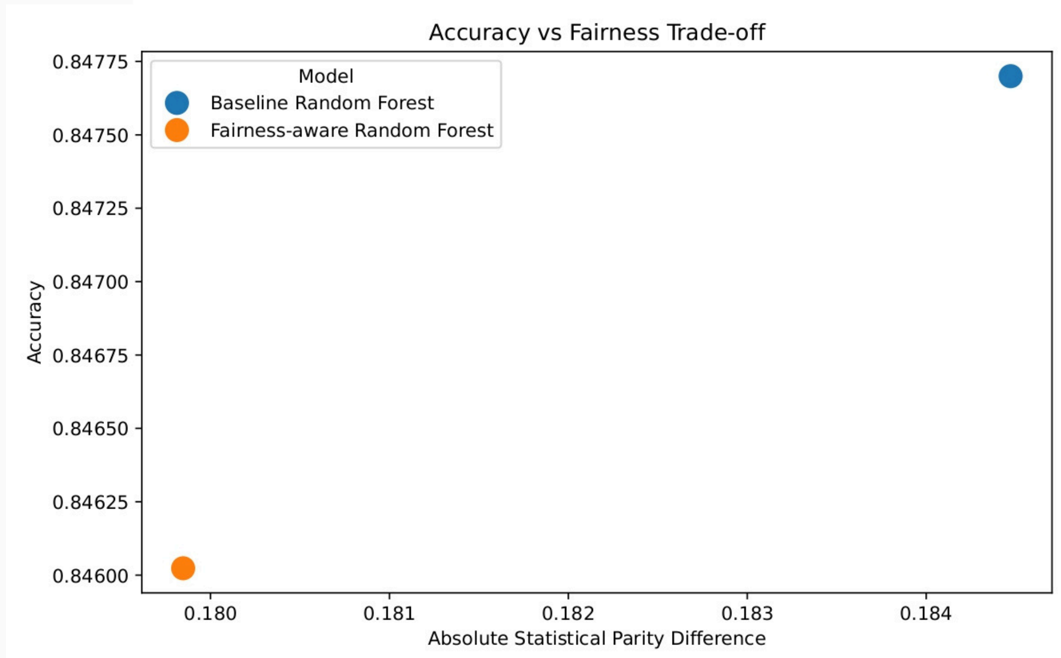
RQ4



RQ2



RQ5



CONCLUSION

- Demographic disparities were identified in AI-based resource allocation.
- Gender influenced the allocation outcomes.
- Race influenced the allocation outcomes.
- Fairness-aware mitigation reduced bias.
- Predictive performance remained largely unchanged.
- Fairness auditing helps promote responsible and equitable AI.